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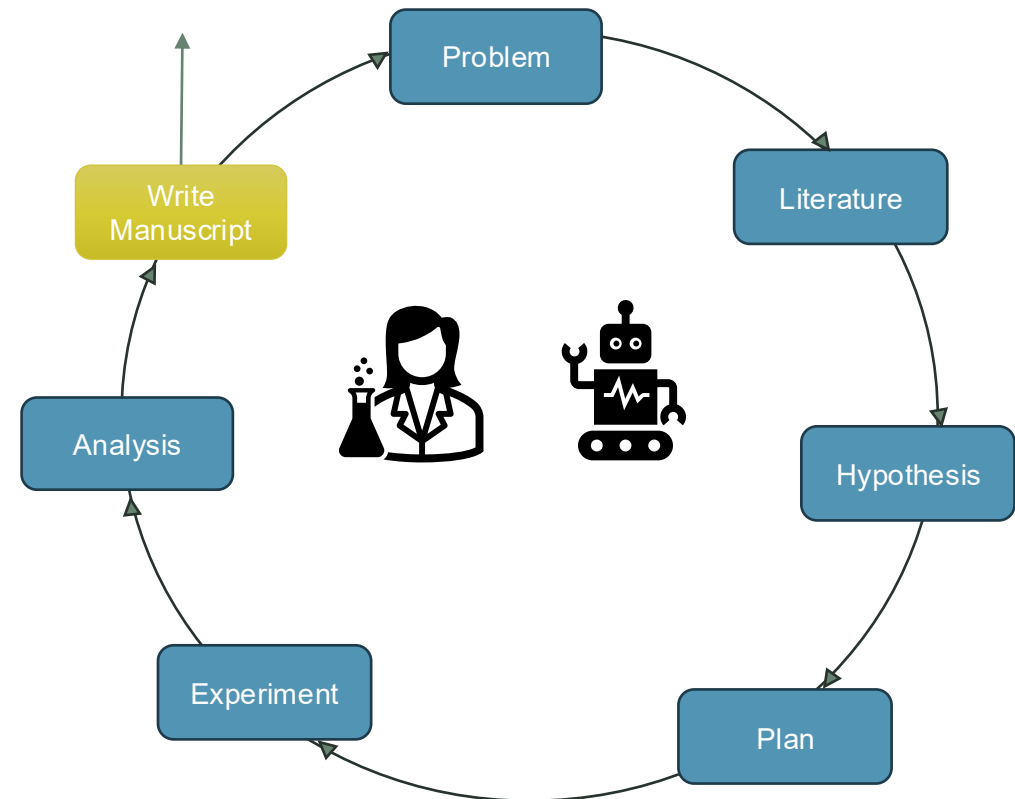
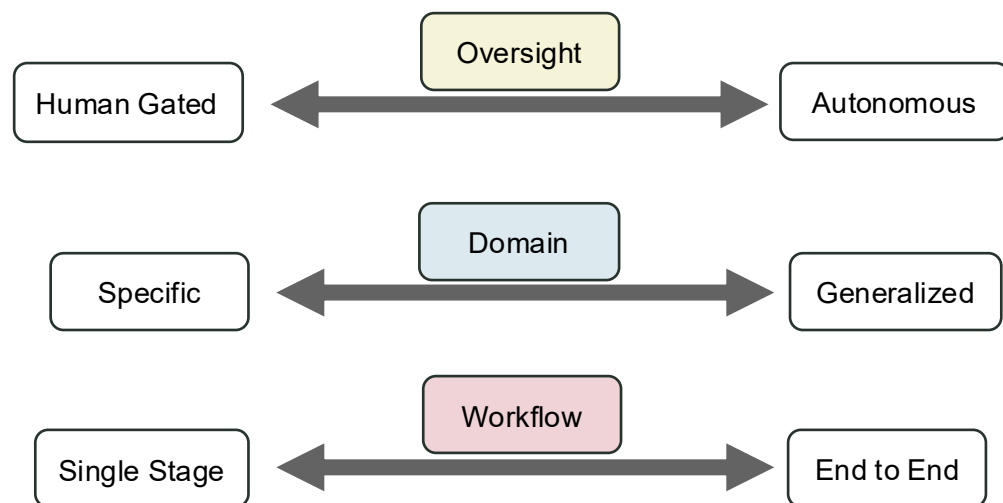
Autonomous Research Systems

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Autonomous Research Systems

Autonomous research systems combine LLMs, tools and orchestration to automate parts or all of the scientific research workflow.



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Autonomous Research Systems

Examples

Domain	2024	2025	2026
Experimental Biology		<u>Co-Scientist (Google) [2]</u> <u>Robin [1]</u>	<u>SciResearcher [6]</u>
Physics			<u>JFC [7]</u>
Aerospace / Engineering			<u>AutoResearch[8]</u>
General / Multidisciplinary		<u>Denario [9]</u> <u>Kosmos [10]</u>	<u>AutoResearchClaw [4]</u>
Machine Learning / AI	<u>The AI Scientist [3]</u> <u>CycleResearcher [17]</u>	<u>Agent Laboratory [12]</u> <u>The AI Scientist-v2 [13]</u> <u>AI-Researcher [14]</u> <u>AIRA [15]</u>	<u>ScientistOne (Google) [11]</u> <u>FARS [16]</u> <u>EurekAgent [5]</u>

*Includes **fully E2E** systems and **partial E2E** systems (automating one or more stages such as hypothesis generation, experimental search or training-data construction)*

[1] Ghareeb et al. (2026), **Robin**, *Nature* · [2] Gottweis et al. (2026), **Co-Scientist**, *Nature* · [3] Lu et al. (2024), **AI Scientist**, arXiv:2408.06292 · [4] Liu et al. (2026), **AutoResearchClaw**, arXiv:2605.20025 · [5] Xin et al. (2026), **EurekAgent**, arXiv:2606.13662 · [6] Zheng et al. (2026), **SciResearcher**, arXiv:2605.01489 · [7] Moreno et al. (2026), **JFC**, arXiv:2603.20179 · [8] Jain et al. (2026), **Agentic AutoResearch**, arXiv:2606.20394 · [9] Villaescusa-Navarro et al. (2025), **Denario**, arXiv:2510.26887 · [10] Mitchener et al. (2026), **Kosmos**, arXiv:2511.02824 · [11] Meng et al. (2026), **ScientistOne**, arXiv:2605.26340 · [12] Schmidgall et al. (2025), **Agent Laboratory**, *EMNLP* · [13] Yamada et al. (2025), **AI Scientist-v2**, arXiv:2504.08066 · [14] Tang et al. (2026), **AI-Researcher**, *NeurIPS* · [15] Toledo et al. (2026), **AIRA**, *NeurIPS* · [16] Tang et al. (2026), **FARS**, arXiv:2606.31651 · [17] Weng et al. (2024), **CycleResearcher**, arXiv:2411.00816

u^b Researchers Optional?

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Accelerating scientific discovery with Co-Scientist

[Juraj Gottweis](#) , [Wei-Hung Weng](#) , [Alexander Daryin](#), [Tao Tu](#), [Petar Sirkovic](#), [Artiom Myaskovsky](#),
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AI-Researcher: Autonomous Scientific Innovation

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Build Your Personalized Research Group: A Multiagent Framework for Continual and Interactive Science Automation

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Dirk Bergemann¹ Zhuoran Yang¹

The Last Human-Written Paper: Agent-Native Research Artifacts

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Workflow Closure Is Not Scientific Closure in Auto-Research Systems

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Position: Correct Answer, Wrong Mechanism — When AI Scientists Defend General Claims Their Own Data Contradicts

Steven Young Eulig¹

The More You Automate, the Less You See: Hidden Pitfalls of AI Scientist Systems

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Perspective | Published: 22 May 2026

AI-induced never-skilling in medical education

[Yuhe Ke](#), [Liyuan Jin](#), [Jasmine Chiat Ling Ong](#), [Arun J. Thirunavukarasu](#), [Josip Car](#), [Carol Y. Cheung](#), [Yih](#)
[Chung Tham](#), [Daniel Shu Wei Ting](#), [Marcus Eng Hock Ong](#), [Scott Compton](#), [Aditee Narayan](#), [Pearse A.](#)
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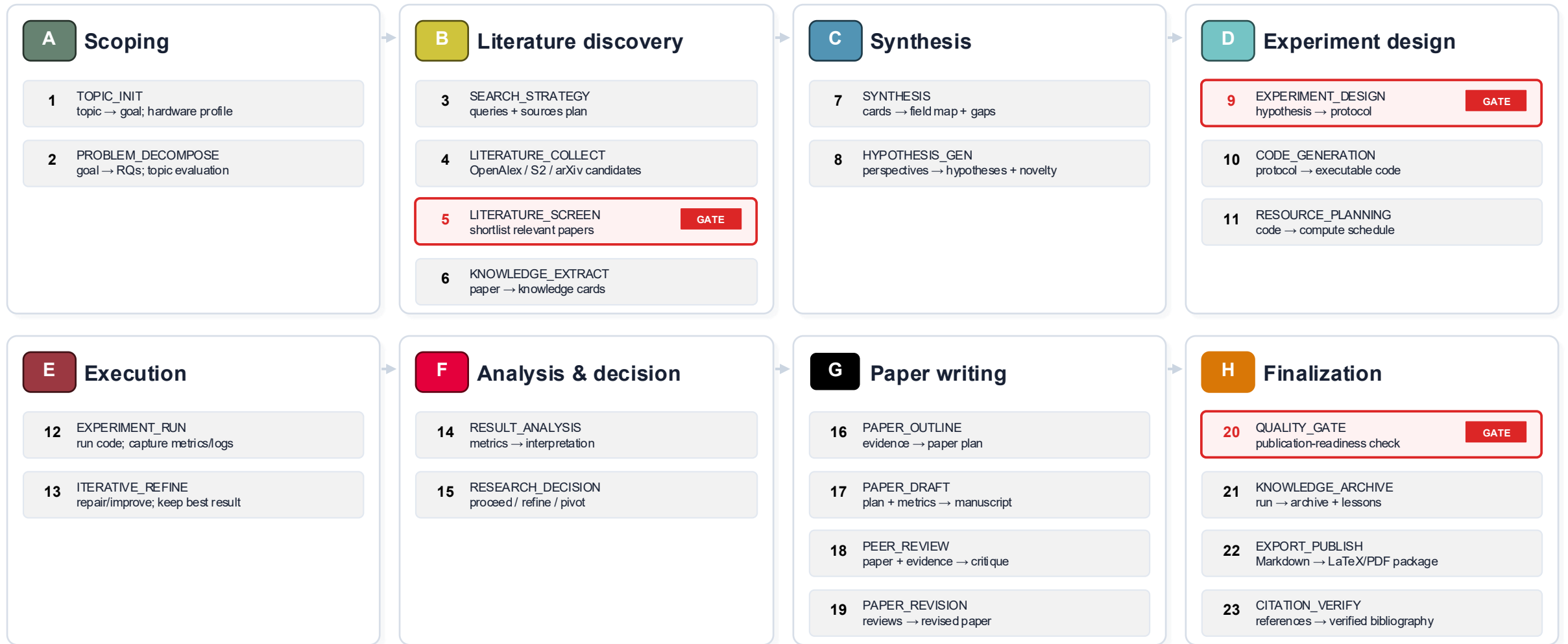
What is Left for Us?

Second Scholarship Against the Degradation of Research by AI

Claudio Novelli^a, Luciano Floridi^{a,b}

u^b AutoResearchClaw

23 Stage Pipeline Summary



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AutoResearchClaw Demo

Github Link : <https://github.com/aiming-lab/AutoResearchClaw>

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Auto Research Claw

Observations From Actual Runs

Good	Bad	Ugly
- Engineering workflow is smooth. - Many different autonomous modes possible (hands off, hands on). - The researcher can put in their inputs (references, goals etc.) in different stages and also intercept via chat.	- Code Implementation can fail and not restart. - Streamed logs are not saved after the run.	- Researcher given references or goals are overwritten. - Accessing search API failed silently and produced partially hallucinated references. - Cannot revisit earlier stages to fix errors propagated downstream

Potentially Interesting Autonomous Research systems for KPM

System	Theme	What is automated?
<u>Project APE [1]</u>	Policy evaluation / economics	Generates empirical policy-evaluation papers end-to-end: ideas, data, econometrics, code, manuscript, revision.
<u>EXPERIGEN [2]</u>	Political/social hypothesis discovery	Generator proposes hypotheses; Experimenter tests over social-science data.
<u>AgentSociety² [3]</u>	E-democracy / executable social science	AI social scientists coordinate literature grounding, hypothesis design, simulations, interpretation and manuscript drafting.
<u>Automated Social Science [4]</u>	Justice, bureaucracy, causal mechanisms	Builds structural causal models, instantiates LLM subjects, designs in-silico experiments and estimates effects.
<u>S-Researcher [5]</u>	Social Sciences	A scale social simulation system for human behavior with LLM agents, analyzing results, and generating reports

[1] **Project APE**, Social Catalyst Lab, UZH · [2] Gupta et al. (2026), **EXPERIGEN**, arXiv:2602.07983 · [3] Piao et al (2026), **AgentSociety 2**, arXiv:2607.11895
[4] Manning et al (2024), Automated Social Science, arXiv:2404.11794 · [5] Wang et al (2026), **S-Researcher**, arXiv:2604.01520

Potentially Interesting Autonomous Research systems Evidence synthesis

System	Domain	Workflow covered	Autonomy	Evidence synthesis	Why interesting
AgentSLR	Epidemiology	Retrieval → screening → PDF/OCR → full-text screening → extraction → report synthesis	High ; end-to-end agentic pipeline, with HITL validation used in evaluation	Narrative/structured synthesis	Open-source and evaluated against expert-curated ground truth across 9 pathogen reviews; particularly useful for studying errors across the whole trajectory. (arXiv)
MedSR-Copilot	Medical systematic reviews	Protocol/PICOS → retrieval → title/abstract/full-text screening → extraction → risk-of-bias → quantitative synthesis → conclusions	High , although designed as a <i>copilot</i> and preserves human-review points	Meta-analysis + final evidence conclusions	Most methodologically complete: explicitly PRISMA-aligned, includes RoB assessment and structured intermediate artifacts. Also has MedSR-Bench for end-to-end evaluation. (MedRxiv)
meta-pipe	Clinical evidence / meta-analysis	Research question → search → screening → extraction → statistical analysis → manuscript	Medium–High ; Claude Code orchestrates the pipeline, with researcher interaction/checkpoints	Meta-analysis using reproducible Python/R tooling	Interesting architecture because it separates LLM reasoning from deterministic statistical analysis rather than asking the model to calculate results itself. (GitHub)
Fully Automated Systematic Review Generation	Medicine	Topic → literature search → inclusion evaluation → extraction → synthesis → manuscript	Very high / fully autonomous ; designed to run without human intervention	Systematic-review synthesis and manuscript generation	Best “extreme autonomy” case: essentially a single-script pipeline. Useful for examining how errors propagate when there are no intermediate human gates . (MedRxiv)

u^b Thanks for your attention!

Questions?

Do not hesitate to get in touch with us if something piqued your curiosity:

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Data Science Lab Walk-In

Do you have trouble running a data processing script?
Do you want to discuss how to use Machine Learning in your project?
Do you have any other question on Data Science or Research Software and IT?

Just come by and we'll try to help you on the spot!

Every Tuesday and Thursday, 13:00–15:00

in the new DSL offices at Uni Muesmatt
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